Software Requirement Specification (SRS) document

A Software Requirements Specification (SRS) is also known as **SRS report**, **software document**.

It is a **document** that

- describes the **nature of a project, software or application,**
- contains description of **all aspects of the software to be built that must be specified before the project is to commence** and
- is also known as **SRS report**, **software document**.

Template for SRS document

These are a **set of guidelines to be followed** while preparing the software requirement specification document.

Different templates are available that can **serve as a guideline** for those who must create a complete SRS.

Contents of SRS template

- purpose
- scope
- functional requirements
- non-functional requirements
- software requirements
- hardware requirements
- environmental conditions required
- safety and security requirements
- software quality attributes of the project and etc.

IEEE Template for SRS

Table of Contents

Revision History

1. Introduction
- 1.1 Purpose
- 1.2 Document Conventions
- 1.3 Intended Audience and Reading Suggestions
- 1.4 Project Scope
- 1.5 References

2. Overall Description
- 2.1 Product Perspective
- 2.2 Product Features
- 2.3 User Classes and Characteristics
- 2.4 Operating Environment
- 2.5 Design and Implementation Constraints
- 2.6 User Documentation
- 2.7 Assumptions and Dependencies

3. System Features
- 3.1 System Feature 1
- 3.2 System Feature 2 (and other features)

4. External Interface Requirements
- 4.1 User Interfaces
- 4.2 Hardware Interfaces
- 4.3 Software Interfaces
- 4.4 Communications Interfaces

5. Other Non-functional Requirements
- 5.1 Performance Requirements
- 5.2 Safety Requirements
- 5.3 Security Requirements
- 5.4 Software Quality Attributes

6. Other Requirements
- Appendix A: Glossary
- Appendix B: Analysis Models
- Appendix C: Issues List

SRS for Library Information System (LIS)

Problem statement for Library Information System (LIS)

- An institute has proposed to **develop a Library Information System (LIS)** for **students and employees.**
- LIS will enable the members **to borrow a book (or return it)** with ease while sitting at his desk/chamber.
- Issuing or returning books is **restricted to valid users** (members) of LIS only.
- The system also enables a member **to extend the date of his borrowing,** if no other booking for that particular book has been made.
- The librarian can **enter a new record into the system when a new book has been purchased**, or **remove a record in case any book is taken off the shelf**.
- Any non-member is free to use this system **to browse/search books online**.
- No password should not be stored in system **as a plain text**.
- The final software would a **web application** (using the recent HTML 5), used **within the institute LAN**.

Requirement identification in Library Information System (LIS)

- Borrow book
- Return book
- Issue book
- Extend the date of his borrowing /re-issue book
- Valid users (members)
- Enter a new record into the system when a new book has been purchased
- Remove a record in case any book is taken off the shelf
- Browse/search books online

Identification of functional requirements

The functional requirements of the system can be:

- **New user/member registration**
- **Search book**
- **User/member login**
- **Issue book**
- **Return book**
- **Re-issue book**
- **Add book**
- **Delete book**

Priority for functional requirements

Having identified the functional requirements, an identifier may be assigned to each of them for **future reference and verification**.

Following table shows the **identifier and priority for software requirements** as

Requirement No.	Requirement	Priority
R1	New user registration	High
R2	User Login	High
R3	Search book	High
R4	Issue book	High
R5	Return book	High
R6	Add book in record	High
R7	Remove book from record	High
R8	Re-issue book	Low

Identification of non-functional requirements

- **Performance Requirements:**
  - This system should remain **accessible 24x7**.
  - At least **50 users** should be able to access the system altogether **at any given time.**
- **Security Requirements:**
  - This system should be accessible **only within the institute LAN**.
  - The database of LIS should not store any password **in plain text - a hashed value has to be stored.**

- **Software Quality Attributes**
    - Reliability
    - Availability
    - Security
    - Portability
- **Database Requirements**
    - File Format (data shall be stored in **text-based flat files**)
    - Accessibility and security
- **Design Constraints:**
    - The LIS has to be developed as a web application, which should work with Firefox 5, Internet Explorer 8, Google Chrome 12.
    - The system should be developed using HTML 5.

Once all the functional and non-functional requirements have been identified, they are **documented formally in SRS, which then serves as a legal agreement for all stakeholders.**